

Multiple Choice (10 marks)

1. Which of the following statements is not true?
 - (a) Humans cannot convert any fatty acids into docosahexaenoic acid
 - (b) Flaxseeds are a rich source of docosahexaenoic acid
 - (c) Docosahexaenoic acid levels are consistently higher in vegans than in omnivores
 - (d) Vegan diets provide an abundance of docosahexaenoic acid naturally
 - (e) Lack of docosahexaenoic acid in vegans causes visual and cognitive impairment
2. How long can a cannula remain in situ?
 - (a) 96 hours.
 - (b) 36 hours.
 - (c) 60 hours.
 - (d) 2 weeks.
 - (e) 1 week.
3. Which are the FIVE main series of apoproteins that have been identified?
 - (a) apoA, apo(a), apoB, apoC and apoH
 - (b) apoA, apoB, apoC, apo E, and apoL
 - (c) apoA, apo(a), apoB, apoC and apoE
 - (d) apoA, apoB, apoC, apoD, and apoF
 - (e) apoA, apoB, apoC, apo(a) and apoM
4. In a condition such as schizophrenia the recurrence risk will be greatest for which of the following relatives of an affected individual?
 - (a) Aunt
 - (b) Nephew
 - (c) Brother
 - (d) Grandchild
 - (e) Great-grandchild
5. Muscle lactate production increases when:
 - (a) the muscle is fully hydrated.
 - (b) oxygen levels in the muscle are low.
 - (c) glucose levels in the muscle are low.
 - (d) oxygen is readily available.
 - (e) the pH of the muscle falls.

True / False (10 marks)

Answer True or False

6. True or False: Biotin plays a crucial role in the carboxylation of pyruvate to form oxaloacetate during gluconeogenesis.
7. True or False: A daily dose of Cotrimoxazole prescribed at 960 mg every 12 hours amounts to 0.96 grams.
8. True or False: Inspecting the nail-bed curvature from below is the most accurate way to evaluate for clubbing.
9. True or False: Grandparents feel particularly cherished when their college-enrolled grandchildren pay surprise visits and seek their advice.
10. True or False: Dental plaque is most likely to accumulate on the buccal surfaces of the upper molars and the lingual surfaces of the lower incisors.

Long Form Response (20 marks)

Answer each question with a clear and complete explanation.

11. Discuss the relationship between optimism and longevity in older adults, analyzing how pessimism may contribute to shorter lifespans and its implications for health.
12. Discuss the role of amino acids in gluconeogenesis and analyze why tryptophan is not a suitable substrate for this metabolic pathway.

End of Paper